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Studierichting: Informatica
Oefeningenreeks 4A nr. 46.1

$$\int \frac{8\left(x^3-1\right)}{9x^4+10x^2+1}dx = 8\int \frac{x^3-1}{9x^4+10x^2+1}dx = \frac{8}{9}\int \frac{x^3-1}{\left(x^2+1\right)\left(x^2+\frac{1}{9}\right)}dx$$

$$\rightarrow \frac{x^3-1}{\left(x^2+1\right)\left(x^2+\frac{1}{9}\right)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+\frac{1}{9}}$$

$$\Rightarrow A\left(i\right)+B = \frac{i^3-1}{i^2+\frac{1}{9}} = \frac{9}{8}+\frac{9}{8}i \Rightarrow A=\frac{9}{8}, B=\frac{9}{8}$$

$$\Rightarrow C\left(\frac{1}{3}i\right)+D = \frac{-\frac{1}{27}i-1}{-\frac{1}{9}+1} = -\frac{9}{8}-\frac{1}{24}i \Rightarrow C=-\frac{1}{8}, D=-\frac{9}{8}$$

$$= \frac{8}{9}\int \frac{\frac{9}{8}x+\frac{9}{8}}{x^2+1}dx + \frac{8}{9}\int \frac{-\frac{1}{8}x-\frac{9}{8}}{x^2+\frac{1}{9}}dx$$

$$= \frac{1}{9}\int \frac{\frac{9}{2}\left(2x\right)+9}{x^2+1}dx + \frac{1}{9}\int \frac{-\frac{1}{2}\left(2x\right)-9}{x^2+\frac{1}{9}}dx$$

$$= \frac{1}{2}\ln\left|x^2+1\right| + \operatorname{Bgtan} x - \frac{1}{18}\ln\left|x^2+\frac{1}{9}\right| - \frac{1}{\frac{1}{3}}\operatorname{Bgtan}\left(\frac{1}{\frac{1}{3}}x\right)+c$$

$$= \frac{1}{2}\ln\left|x^2+1\right| + \operatorname{Bgtan} x - \frac{1}{18}\ln\left|x^2+\frac{1}{9}\right| - 3\operatorname{Bgtan}\left(3x\right)+c$$

References